5 Solutions to Power Question

1. The values are as follows.
The value of $p_2(4)$ is **4** from $(4,0,0,0)$, $(2,2,0,0)$, $(2,1,1,0)$, and $(1,1,1,1)$.
The value of $p_2(5)$ is **4** from $(4,1,0,0,0)$, $(2,2,1,0,0)$, $(2,1,1,1,0)$, and $(1,1,1,1,1)$.
The value of $p_2(6)$ is **6** from $(4,2,0,0,0,0)$, $(4,1,1,0,0,0)$, $(2,2,2,0,0,0)$, $(2,2,1,1,0,0)$, $(2,1,1,1,1,0)$, and $(1,1,1,1,1,1)$.
The value of $p_2(7)$ is **6** from $(4,2,1,0,0,0,0)$, $(4,1,1,1,0,0,0)$, $(2,2,2,1,0,0,0)$, $(2,2,1,1,1,0,0)$, $(2,1,1,1,1,1,0)$, and $(1,1,1,1,1,1,1)$.

2. **a.** A binary partition for n can be changed into a binary partition for $n+1$ by inserting a 1 immediately following the rightmost nonzero entry. For example, $(2,2,1,1,0,0)$ is a binary partition of 6, and this can be changed into $(2,2,1,1,1,0,0)$, which is a binary partition of 7. Similarly, $(1,1,1,1)$ becomes $(1,1,1,1,1)$, and these are binary partitions of 4 and 5, respectively. Different binary partitions of n become different binary partitions of $n+1$ in this way. Because every binary partition of n maps to a binary partition of $n+1$, there are at least as many binary partitions of $n+1$ as of n.

b. The answer to the question is **no**. If n is even, then $n+1$ is odd. Because a binary partition of an odd number must contain at least one 1, the correspondence described in the solution to Problem 2a is one-to-one. To establish this, given a binary partition of $n+1$, delete one 1 to obtain a binary partition of n. Thus $p_2(n+1) = p_2(n)$ for any even n. However, note that if n is odd, then $p_2(n+1)$ is strictly greater than $p_2(n)$ because in addition to the binary partitions of $n+1$ that can be obtained by inserting a 1 into a binary partition of n, there are binary partitions of $n+1$ consisting of all even numbers.

3. **a.** Let n be an even integer. Given a binary partition of n, either all of its parts are even or there is at least one 1. If all parts are even, then all of the positive parts must occur in the first $\frac{n}{2}$ elements of the n-tuple (otherwise, their sum would exceed n). Dividing all the parts by 2 yields an n-tuple of powers of 2 and 0s that sum to $\frac{n}{2}$. Note that the last $\frac{n}{2}$ elements of this n-tuple must all be 0s and truncating them results in an $\left(\frac{n}{2}\right)$-tuple which is a binary partition of $\frac{n}{2}$. Conversely, for any binary partition of $\frac{n}{2}$, by doubling each element and appending $\frac{n}{2}$ zeros at the end results in an n-tuple which is a binary partition of n. Hence $p_2\left(\frac{n}{2}\right)$ counts the number of binary partitions of n in which all the parts are even. If there is at least one 1, then the assumption that n is even means that there are at least two 1s, and deleting them produces a binary partition of $n-2$. Conversely, for each binary partition of $n-2$, appending two 1s (right before the 0s, if there are any, otherwise, append them to the end of the $(n-2)$-tuple) produces an n-tuple that is a binary partition of n. Hence $p_2(n-2)$ counts the number of binary partitions of n with at least one 1, and thus the desired recursion follows.

b. **There is no such n.** Proceed by cases. If n is odd, then $p_2(n) = p_2(n-1)$ because all binary partitions of an odd number include at least one 1, and removing that 1 gives a bijection to partitions of $n-1$. If n is even, then $p_2(n) = p_2\left(\frac{n}{2}\right) + p_2(n-2)$ by Problem 3a. Because $p_2(2) = 2$ and $p_2(4) = 4$, it follows inductively that $p_2(n)$ is even for all $n > 1$.

4. **a.** The four binary partitions of 5 are ordered in the following way: $(1,1,1,1,1) \prec (2,1,1,1,0) \prec (2,2,1,0,0) \prec (4,1,0,0,0)$. This is a total ordering.

b. Consider these two binary partitions of 8: $(4,1,1,1,1,0,0,0)$ and $(2,2,2,2,0,0,0,0)$. The 4 can break down into two 2s, but that is the only way to generate 2s, and that does not generate four 2s, so it is not the case that $(2,2,2,2,0,0,0,0) \prec (4,1,1,1,1,0,0,0)$. Likewise, it is not possible for a partition with no 4s to be related by $\prec$ to a partition with 4s, so it is not the case that $(4,1,1,1,1,0,0,0) \prec (2,2,2,2,0,0,0,0)$.

Similarly, the following two pairs of partitions are not comparable:

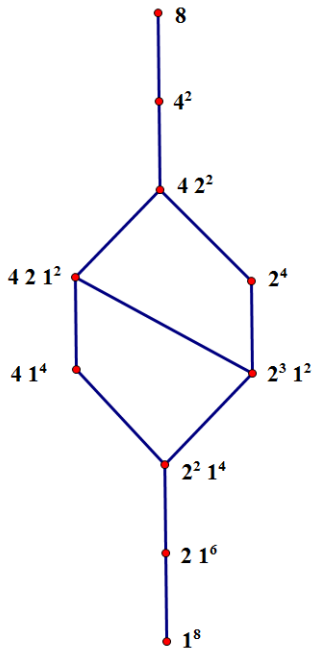
$$(4,1,1,1,1,0,0,0) \text{ and } (2,2,2,1,1,0,0,0), \qquad (4,2,1,1,0,0,0,0) \text{ and } (2,2,2,2,0,0,0,0).$$

5. **a.** The smallest binary partition of n is the partition with n 1s. That is, it is the partition $(1,1,\ldots,1)$ with n copies of 1 in the partition. If a partition has any part that is not equal to 0 or 1, then that part is a positive power of 2 that may be replaced by lesser powers of 2. This process can continue until all of the powers of 2 are 1s, but the process cannot continue beyond that because a partition must consist only

of integers. Thus every binary partition is greater than the one consisting of all 1s. Also note that the partition $(1, 1, \ldots, 1)$ is not larger than or incomparable with any other partition. Thus $(1, 1, \ldots, 1)$ is the smallest binary partition.

b. The largest binary partition of n is the one that essentially gives the base-2 equivalent of n. That is, if the base-2 representation of n has 1s in the 2^a, 2^b, 2^c places, and so on, with $a > b > c > \cdots$, then the largest binary partition of n is $(2^a, 2^b, 2^c, \ldots, 0, 0, 0)$ (with enough 0s to finish the partition). If a partition contains two copies of a number other than 0, they may be combined to form a greater power of 2. Thus, starting with any partition, continue combining powers of 2 until the partition contains no repeated numbers. Because no further combinations are allowed at this point, by the definition, there can be no partition larger than a partition with no repeated numbers. *A priori* there could be other partitions that are not comparable to such a partition, however. But because such a partition expresses n as a sum of powers of 2, with no repeated nonzero parts, it is essentially the base-2 representation of n, which is unique. So because any partition may undergo the process of combining like powers of 2 until there are no repeats, every binary partition is less than the unique partition which expresses n in binary.

6. **a.** A Hasse diagram for the binary partitions of 8 is shown below.



b. The following sequence of vertices is a Hamiltonian path: $1^8, 21^6, 2^21^4, 41^4, 421^2, 2^31^2, 2^4, 42^2, 4^2, 8$. This path begins at the vertex at Level 0, then visits each vertex exactly once.

c. The bijection was effectively established in the solution to 3a. Now consider whether $B(p) \prec B(p')$ for p and p' which are different binary partitions of n and for which $p \prec p'$. Suppose first that both p and p' have no 1s at all. Then dividing all parts by 2 preserves the covering relation. This is because p can be produced by swapping out powers of 2 in p' for lesser powers of 2, and dividing all parts by 2 does not change this covering. Now suppose that p and p' both have at least two 1s. Then removing two 1s from each partition preserves the covering relation. This is because removing two 1s from the partition in no way changes the "swap-outs" of powers of 2 for lesser powers of 2. Now suppose that one of p or p' has two 1s and the other doesn't. Then $B(p)$ is incomparable with $B(p')$, because $B(p) \in S_1$ and $B(p') \in S_2$ or vice versa, and these are sets of partitions of different numbers because $n - 2 > \frac{n}{2}$ when $n > 4$. The relation $\prec$ is not defined between partitions of different numbers, so $B(p)$ and $B(p')$ are not comparable.

d. As in the solutions to Problems 2 and 3, it is only necessary to consider the problem for even n. This is because a Hasse diagram for partitions of $2r + 1$ is the same as that for the partitions of $2r$ with an extra 1 in every partition. So let $n = 2r = a \cdot 2^k$, where a is an odd integer and k is positive. The proof proceeds by strong induction to prove that there is a Hamiltonian path through the Hasse diagram for

partitions of n starting at 1^n and ending at $(2^k)^a$. To get started, the path $1^2 \to 2$ is clearly a Hamiltonian path through the vertices of the Hasse diagram for partitions of 2, and the path $1^4 \to 21^2 \to 2^2 \to 4$ is a Hamiltonian path through the vertices of the Hasse diagram for partitions of 4. These establish the base case.

The idea of the proof will be to use the bijection from part c. In essence, the partitions of n will be split into those that are matched with partitions of $n-2$ and those that are partitions of $n/2$, and the Hamiltonian paths through each of these sets of smaller partitions will be joined into a single path through the partitions of n.

The path starts at 1^n. Consider this as $1^{n-2}1^2$. That is, group the first $n-2$ 1s together and the last two 1s together. From the induction hypothesis, there is a Hamiltonian path through the partitions of $n-2$ starting with 1^{n-2}. By adjoining two 1s to each of these, there is a Hamiltonian path through the partitions of n that have two or more 1s in them. This path ends at $(2^\ell)^c 1^2$, where $n-2 = 2(r-1) = c \cdot 2^\ell$, c is an odd integer, and ℓ is positive. The manner in which the path will be continued through the partitions of n with no 1s in them is dependent on the parity of r.

First consider the case in which r is even. Then $r-1$ is odd. In this case, $c = r-1$ and $\ell = 1$. Thus the path in the previous paragraph ended at $2^c 1^2$. The next step in the path will be to $2^{c+1} = 2^r$. Now by induction, there is a Hamiltonian path through the partitions of $n/2 = r$ starting at 1^r and ending at $(2^{k-1})^a$. Double every entry of each of these partitions. This gives a Hamiltonian path through the partitions of n that have no 1s in them, starting at 2^r and ending at $(2^k)^a$. Attaching this to the end of the path from 1^n to 2^r already obtained results in the desired Hamiltonian path from 1^n to $(2^k)^a$.

In the case where r is odd, $r-1$ is even. In this case, $a = r$ and $k = 1$, while $n-2 = 2(r-1)$ must be a multiple of 4. So the path through the partitions of n containing 1s ended at $(2^\ell)^c 1^2$ for some $\ell \geq 2$. The next step will be to the partition $(2^\ell)^c 2$. Now $r = n/2$ is odd, so every binary partition of r must contain at least one 1. Note that these are the binary partitions of $r-1$ with a 1 adjoined, and by induction, there is a Hamiltonian path through them starting at 1^r and ending at $(2^{\ell-1})^c 1$. Double each entry of each of these, and run through this path *backward* to obtain a path starting at $(2^\ell)^c 2$ and ending at 2^r. Concatenating this with the path from 1^n to $(2^\ell)^c 2$ previously obtained will yield the desired path from 1^n to $2^r = (2^k)^a$.

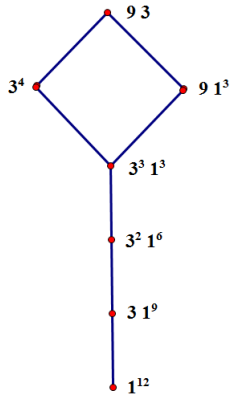
7. Proceed by induction on L. The only element for which $L = 0$ is the smallest binary partition of n, which indeed has n 1s by the solution to Problem 5, and so this partition has $n - L = n - 0$ parts. Moving up via a covering relation, when two 2^ks are replaced by one 2^{k+1}, it is true that one nonzero part is lost. Assuming an element at level L has $n-L$ nonzero parts, an element at level $L+1$ will have $n-L-1 = n-(L+1)$ parts. This establishes the inductive step.

8. **a.** There are several s-partitions of 7. They include $(7,0,0,0,0,0,0)$, $(3,3,1,0,0,0,0)$, $(3,1,1,1,1,0,0)$, and $(1,1,1,1,1,1,1)$. Other partitions of 7 are not s-partitions; some of those are also not binary. They include $(6,1,0,0,0,0,0)$, $(5,2,0,0,0,0,0)$, $(5,1,1,0,0,0,0)$, $(3,2,2,0,0,0,0)$, and $(3,2,1,1,0,0,0)$.

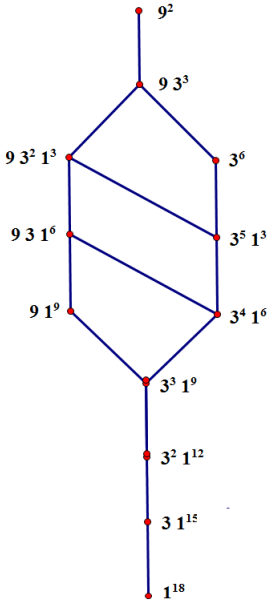
b. Construct a bijection. Given a binary partition of n at level L, subtract 1 from each nonzero part. Hence each part 2^j is replaced by $2^j - 1$, leaving an s-partition of $n-(n-L) = L$.

For the inverse map, consider an s-partition of L, with parts of the form $2^j - 1$ for some positive integer j. This cannot have more than L parts, and that is achievable only with all parts equal to 1. Consider the partition to end in enough 0s to bring the total number of parts up to $n-L$; this is possible because $n - L = n - 2L + L \geq 2L - 2L + L = L$. Now add 1 to each part. The result is a binary partition of $L + n - L = n$ with $n-L$ parts, which puts it at level L by Problem 7.

9. A Hasse diagram for the trinary partitions of 12 is shown below.



10. The three possible answers are **18** or **19** or **20**. A Hasse diagram for the trinary partitions of 18 is shown below. The answer will be established when it is shown that there is no Hamiltonian path through the Hasse diagram. Note that all Hasse diagrams for the trinary partitions of 18 are isomorphic to the one in the figure. Note also that the Hasse diagram for the trinary partitions of 19 is isomorphic to the Hasse diagram for the trinary partitions of 18; a bijection can be established by adding a 1 after the rightmost nonzero part of any trinary partition of 18. Similarly, the Hasse diagram for the trinary partitions of 20 is isomorphic to the Hasse diagram for the trinary partitions of 18.



Suppose there is a Hamiltonian path through the Hasse diagram. Because the Hasse diagram has only one edge coming from 1^{18} and only one edge coming from 9^2, the Hamiltonian path must begin and end at those vertices in some order. Without loss of generality, assume the path begins at 1^{18} and ends at 9^2.

The path must be of the form $1^{18}, 31^{15}, 3^21^{12}, 3^31^9, \ldots, 9^2$. The first choice comes when deciding if the fifth vertex in the path is 91^9 or 3^41^6. If the choice is to make 3^41^6 the fifth vertex in the path, then it is not possible to enter and leave 91^9 because the edge $\{3^31^9, 91^9\}$ cannot be used for entry or for exit; both would require revisiting 3^31^9. So the fifth vertex in the path must be 91^9, and thus the sixth vertex in the path is 931^6.

The next choice comes when considering the seventh vertex in the path. If the seventh vertex is 93^21^3, then it will be impossible to visit 3^41^6 and also 9^2. If the seventh vertex is 3^41^6, then the eighth vertex in the path is 3^51^3, and either choice from that vertex (93^21^3 or 3^6) leads to a contradiction because to visit the

other requires a path that prevents visiting 9^2. This completes the proof that the Hasse diagram of the trinary partitions of 18 does not contain a Hamiltonian path.

Checking other possible values of n that are multiples of 3, there is a Hamiltonian path through the Hasse diagrams for each one, as the following argument demonstrates.

- A Hamiltonian path through the Hasse diagram for $n = 3$ is $1^3, 3$.
- A Hamiltonian path through the Hasse diagram for $n = 6$ is $1^6, 31^3, 3^2$.
- A Hamiltonian path through the Hasse diagram for $n = 9$ is $1^9, 31^6, 3^21^3, 3^3, 9$.
- A Hamiltonian path through the Hasse diagram for $n = 12$ is $1^{12}, 31^9, 3^21^6, 3^31^3, 91^3, 93, 3^4$.
- A Hamiltonian path through the Hasse diagram for $n = 15$ is $1^{15}, 31^{12}, 3^21^9, 3^31^6, 91^6, 931^3, 93^2, 3^5, 3^41^3$.
- A Hamiltonian path through the Hasse diagram for $n = 21$ is $1^{21}, 31^{18}, 3^21^{15}, 3^31^{12}, 91^{12}, 931^9, 3^41^9, 3^51^6$, $93^21^6, 93^31^3, 93^4, 9^23, 9^21^3$.

Note that there are Hamiltonian paths through the Hasse diagram for both $n = 1$ (the path is just the vertex 1) and $n = 2$ (the path is $1^2, 2$). Note also that if k is a positive integer, the Hasse diagrams for the trinary partitions of $n = 3k$, $n = 3k + 1$, and $n = 3k + 2$ are all isomorphic, in the same way that the Hasse diagrams for the trinary partitions of 18, 19, and 20 are isomorphic.

Thus the only values of n less than 23 for which the Hasse diagram of trinary partitions does not contain a Hamiltonian path are 18 or 19 or 20.